

1. (Fall 2008, #3) You want to approximate the integral $\int_1^2 e^{x^2} dx$ by using the Trapezoidal Rule with n intervals. How large should you take n to guarantee that the error is at most 10^{-6} ? (Do not compute the approximation.)

2. (Fall 2008, #3) We want to approximate the value of the integral $\int_1^3 \frac{1}{x^2} dx$ by using the Midpoint Rule and subdividing the $[1, 3]$ into n intervals of equal length. How large should we take n if we want the error to be less than 0.0002?

3. (Stewart, §6.5, 31)

- (a) Use the Midpoint Rule and the given data to estimate the value of the integral $\int_1^5 f(x)dx$.

x	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
$f(x)$	2.4	2.9	3.3	3.6	3.8	4.0	4.1	3.9	3.5

- (b) If it is known that $-2 \leq f''(x) \leq 3$ for all x , estimate the error involved in the approximation in part (a).

4. (Stewart, §6.5, 30)

- (a) A table of values of a function g is given. Use Simpson's Rule to estimate $\int_0^{1.6} g(x)dx$.

x	0.0	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6
$g(x)$	12.1	11.6	11.3	11.1	11.7	12.2	12.6	13.0	13.2

- (b) If $-5 \leq g^{(4)}(x) \leq 2$ for $0 \leq x \leq 1.6$, estimate the error involved in the approximation in part (a).

5. (Stewart, §6.5, 17)

(a) Find the approximations T_8 and M_8 for the integral $\int_0^1 \cos(x^2) dx$.

(b) Estimate the errors in the approximations of part (a).

(c) How large do we have to choose n so that the approximations T_n and M_n to the integral in part (a) are accurate to within 0.0001?